

Piyush Hinduja

Salt Lake City, UT | (385) 507-6190 | piyushinduja20@gmail.com | [Portfolio](#) | [LinkedIn](#)

EDUCATION

Master of Science in Computer Science

University of Utah

GPA: **3.91**/4.0

2023 – 2025

Bachelor of Engineering in Computer Engineering

University of Mumbai

GPA: **3.85**/4.0

2019 – 2023

SKILLS

Languages: *Expert (5+ yrs):* Python, Java, JavaScript *Intermediate (3+ yrs):* C, C++ *Beginner (1+ yrs):* PHP, Kotlin, SQL, Shell

AI/ML: LLMs, PyTorch, HuggingFace, Langchain, TensorFlow, Scikit-Learn, EasyOCR, spaCy

Data Processing: RESTful APIs, Pandas, PySpark, Databricks, OCRs, Matplotlib, Seaborn, DuckDB, PowerBI, Tableau

Applications and Databases: FastAPI, Streamlit, Bash, Preswald, MongoDB, MySQL, Amazon RDS, Firebase, ChromaDB

Cloud & Miscellaneous: AWS (S3, SageMaker, Texteract, IAM), Azure, Docker, Kubernetes, Git, CI/CD, Agile/SCRUM, JIRA

WORK EXPERIENCE

AI Engineer (formerly Data Analyst)

University of Utah

March 2024 – Present

Salt Lake City, Utah

- Building language models to analyze Earnings Conference Calls and 10-K, 10-Q, and 8-K filings, identifying correlations and executive statements linked to future performance trends
- Implemented advanced NLP analysis using **encoder-only model FinBERT, PyTorch, and Transformers** for financial document processing, **improving the prediction accuracy from 66% to 94.13%**
- Architected a robust **ETL pipeline** to extract, parse, and store financial documents from SEC.gov and EDGAR website; optimized data processing using multi-core and multi-threaded techniques, **reducing per-filing processing time by more than 90%**.
- Developed robust **prompt engineering** solutions to integrate LLMs & GenAIs (OpenAI, Anthropic, and Gemini) for automated financial text analysis, comparing AI-generated insights against traditional algorithms

Software Developer

Squeeze Media LLC

May 2025 – June 2025

Orem, Utah

- Built an AI-powered warm transfer system for inbound/outbound **Twilio calls using FastAPI, Deepgram SST, and WebSocket** streaming
- Increased company sales by 28%** by ensuring more customers completed their calls without drop-offs or repeated queries

Machine Learning Researcher

FLUX Research Group

August 2024 – April 2025

Salt Lake City, Utah

- Engineered automated deployment system using **Python and Linux Bash to streamline NVIDIA/CUDA toolkit** integration on high-performance computing clusters, reducing overhead by **90%**
- Analyzed wireless spectrum data to uncover scientific insights and **optimize power distribution across 20+ campus routing devices**

Full Stack Developer Intern

Exposys Data Labs

June 2021 – July 2021

Bangalore, India

- Spearheaded the launch of a tourism project, integrating key features to **enhance user experience** and deliver valuable information about 50+ Indian travel destinations
- Utilized **MERN stack** for an efficient frontend and backend integration, creating a responsive, user-friendly interface and enabling content management

PROJECTS

SEC Comment Letter Analysis | LLMs, OCR, Claude API

- Engineered a rolling-window algorithm to process and bundle SEC comment letter conversations, analyzing data from **50,000+ companies** via the EDGAR database
- Designed an **OCR pipeline using the EasyOCR library** to extract text from PDFs with **95% accuracy**, processing 2-page documents in under 2 seconds
- Constructed prompt-based labeling system using **Claude API** to classify letter content with **98.25% accuracy**, integrating metadata for contextual inference

Legal AI Assistant | LLaMA3, RAG, AWS

- Fine-tuned **LLaMA 3** model on Indian court data extracted via Indian Kanoon and Bharat Law APIs for domain-specific legal reasoning
- Integrated the model into a **RAG-based SaaS platform** that retrieves and cites relevant cases during chat; projected to boost answer accuracy from **56% to 93%**, significantly outperforming generic OpenAI models

Road Damage Detection | Object-Detection, YOLOv5, FasterRCNN

[GitHub Link](#)

- Collaborated with a team of four to develop a road damage detection system using **YOLOv5 and FasterRCNN**, integrating a user-friendly web interface (deployed via **Streamlit**), leveraging the RDD-2020 dataset
- Achieved model accuracies of **52.67% with YOLOv5 and 44.45% with FasterRCNN** by training and fine-tuning on different hyper-parameters
- The system plays a vital role, delivering a **70% improvement in road damage detection accuracy** over manual methods, helping lower the frequency of accidents and fatalities

Dependency Parser | PyTorch, GloVe Embeddings

[GitHub Link](#)

- Architected an end-to-end **Transition-Based Dependency Parser in PyTorch**, implementing a multi-class classifier for predicting parsing actions based on stack and buffer states
- Designed modular pipeline incorporating data preprocessing, model architecture, training workflow, and evaluation metrics
- Integrated **GloVe word embeddings** for enhanced semantic representation, contributing to **UAS of 0.76 and LAS of 0.705**